

# Skills Summary

## Reading and Comparing Multi-Digit Numbers

*Use this reference while completing your worksheet or when studying.*

### Skill 1 — The Value of a Digit

**Rule:** value of a digit = digit  $\times$  its place. Each place is worth **10 times** the place to its right: ten-thousands  $\cdot$  thousands  $\cdot$  hundreds  $\cdot$  tens  $\cdot$  ones.

**Example:** Find the value of the 7 in 74,206.

**Step 1.** Locate the place first: counting from the ones, the 7 sits in the ten-thousands place, so multiply the digit by 10,000.

**Step 2.**  $7 \times 10,000 = 70,000 \Rightarrow$  **70,000**

**Try it:** Find the value of the 3 in 5,382.

check: 3000

### Skill 2 — Number Forms

**Three ways to write one number:** standard form (24,601), expanded form (a sum of place values), and word form. An empty place gets a **zero** in standard form.

**Example:** Write  $20,000 + 4,000 + 600 + 1$  in standard form.

**Step 1.** Each addend names one place, so fill the places from largest to smallest and put a zero in any place with no addend.

**Step 2.** 2 ten-thousands, 4 thousands, 6 hundreds, 0 tens, 1 one:  $20,000 + 4,000 + 600 + 1 = 24,601 \Rightarrow$  **24,601**

**Try it:** Write  $50,000 + 800 + 20 + 7$  in standard form.

check: 50,827

### Skill 3 — Comparing Numbers

**Rule:** (1) More digits wins. (2) Same number of digits: line up the places and compare from the **left**; the first place that differs decides. Symbols:  $<$  less than,  $>$  greater than,  $=$  equal.

**Example:** Compare 36,254 and 36,245.

**Step 1.** Both have five digits, so line up the places and scan from the left until a place differs.

**Step 2.** Ten-thousands  $3 = 3$ , thousands  $6 = 6$ , hundreds  $2 = 2$ ; tens differ:  $5 > 4. \Rightarrow$  **36,254  $>$  36,245**

**Try it:** Compare 8,109 and 8,190 with  $<$ ,  $>$ , or  $=$ .

check:  $8109 < 8190$

**(!) Watch out:** Never compare just the first digits. 9,876 starts with a 9, but it is *smaller* than 12,345 — the five-digit number wins because it reaches a higher place.